

# The symmetric Mahler conjecture and its equality cases

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## Abstract

We resolve the symmetric Mahler conjecture positively, including its equality classification. Every origin-symmetric convex body in  $\mathbb{R}^n$  has volume product at least  $4^n/n!$ , with equality exactly for invertible linear images of Hanner polytopes.

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## 1 Introduction

A convex body is a compact convex subset of  $\mathbb{R}^n$  with nonempty interior. If  $K = -K$ , its center 0 is an interior point, and its polar body is

$$K^\circ = \{y \in \mathbb{R}^n : \langle x, y \rangle \leq 1 \text{ for every } x \in K\}.$$

Write  $|K|$  for  $n$ -dimensional Lebesgue volume and  $P(K) = |K| |K^\circ|$  for the volume product. The two determinant factors cancel under an invertible linear change of coordinates, so  $P(TK) = P(K)$ . Mahler's symmetric volume-product problem asks how small this linear invariant can be. The conjectured value is  $4^n/n!$ , the product of a cube and its polar cross-polytope. We prove this bound in every dimension and classify the equality cases.

The extremal bodies are defined recursively. A centered nondegenerate interval is a one-dimensional *Hanner polytope*. Given Hanner polytopes  $H_1 \subset \mathbb{R}^k$  and  $H_2 \subset \mathbb{R}^l$ , their Cartesian product and the convex hull

$$\begin{aligned} H_1 \oplus_\infty H_2 &= H_1 \times H_2, \\ H_1 \oplus_1 H_2 &= \text{conv}((H_1 \times \{0\}) \cup (\{0\} \times H_2)) \end{aligned}$$

are Hanner polytopes in  $\mathbb{R}^{k+l}$ . These are the  $\ell_\infty$  and  $\ell_1$  sums, respectively. This class contains cubes, cross-polytopes, and mixed recursive sums. We call an invertible linear image of a Hanner polytope a *linear Hanner body*.

**Theorem 1.1.** *For every integer  $n \geq 1$  and every origin-symmetric convex body  $K \subset \mathbb{R}^n$ ,*

$$|K| |K^\circ| \geq \frac{4^n}{n!}. \quad (1)$$

*Equality holds if and only if  $K$  is a linear Hanner body.*

The linear equivalence in the statement respects the fixed origin used for polarity.

## 1.1 History and context

Mahler introduced the volume-product problem in his work on convex bodies and transference in the geometry of numbers [22]; the planar inequality also goes back to him [21]. Reisner later characterized its equality cases as parallelograms, as a consequence of his zonoid theorem [28, p. 340]. Iriyeh and Shibata proved the complete three-dimensional symmetric theorem, including equality [14, Theorem 1.1]. Fradelizi, Hubard, Meyer, Roldán-Pensado, and Zvavitch gave a shorter proof based on equipartitions, together with another equality argument and stability [9]. Chen, Li, Xi, and Xu recently gave a shadow-flow proof of that theorem in a preprint that also treats the nonsymmetric three-dimensional problem [4, Theorem 7.1].

Hanner's recursively constructed spaces [12] already play a central role in the known higher-dimensional cases. Saint-Raymond proved the sharp inequality for unconditional bodies (those invariant under coordinate sign changes) [30]; Meyer and Reisner identified the Hanner equality cases in that class [23, 29]. Reisner also proved the sharp inequality for zonoids, which are Hausdorff limits of Minkowski sums of centered segments, and showed that equality within that class means a parallelotope [27, 28]; Gordon, Meyer, and Reisner later gave a short geometric proof [10]. For local questions in Banach–Mazur distance, Nazarov, Petrov, Ryabogin, and Zvavitch established strict local minimality, up to linear equivalence, of the cube [25]. Kim extended this to all Hanner polytopes [16]. Kim and Zvavitch obtained stability in the unconditional class and the inequality in a neighborhood of that class [17].

A separate line of work seeks dimension-independent exponential bounds without the sharp constant. Bourgain and Milman established such a reverse Blaschke–Santaló inequality [3]; Kuperberg obtained an explicit stronger bound by a geometric argument using Gauss linking integrals [18]. Complex-analytic approaches provide another setting for these questions. Nazarov used Hörmander's  $\bar{\partial}$  theorem and Bergman kernels for the reverse Blaschke–Santaló inequality [24]. Berndtsson recast

Kuperberg’s proof using complex integrals [1]. In a symplectic approach, Karasev proved the sharp inequality for hyperplane sections and projections of  $\ell_p$  balls ( $1 \leq p \leq \infty$ ) and Hanner polytopes [15, Theorems 4.2 and 4.4]. These works give analytic and geometric precedents for volume-product estimates; their estimates are not hypotheses of our proof.

The analytic ingredient is a conformal map of the disk for which uniform angular measure on the boundary maps to a uniform imaginary coordinate. It is a rotation of Gross’s example for the uniform distribution in his conformal Skorokhod embedding [11, Section 3.2]. A holomorphic mass estimate turns that boundary law into a lower bound for an integral of feasible-simplex volumes over the primal body. We prove the mass estimate directly by Stokes’ theorem, with its precise normalization; its general background is the theory of generalized Lelong numbers [6, Sections 3 and 5]. The symmetric-convex extremal-function framework of Lundin and Baran was a further motivation for the analytic search; see [20] and the formula recorded in [2, Proposition 1.2, pp. 246–247]. The proof here instead uses the direct isolated-zero mass calculation, not those extremal-function formulas as premises. The equality argument leads to a classical property of normed spaces: any three pairwise-intersecting closed balls have a common point. Hanner studied the associated recursively constructed convex bodies [12]. Lima’s norm-additive decomposition criterion [19, Theorem 1.2], also recorded in Hansen–Lima [13, Theorem 3.6(4)], is exactly the metric-median condition used below. Reisner’s unconditional equality theorem already connects minimal volume product to this ball intersection structure [29, Theorem 1]. Our endpoint argument obtains the structure without assuming unconditionality; Hansen–Lima’s finite-dimensional classification is the final input [13, Corollary 7.4].

For a body without a prescribed center, the general Mahler problem instead minimizes  $|K| |(K - z)^\circ|$  over  $z \in \text{int } K$ . Its conjectured constant  $(n+1)^{n+1}/(n!)^2$  is smaller than the symmetric constant for  $n \geq 2$ ; a result for that general problem alone therefore does not establish Theorem 1.1. The argument here uses symmetry throughout and is independent of the general theorem proved in the companion paper [26, Theorem 1.1].

## 1.2 Proof overview

Section 2 records polarity and the two norm-sum formulas. They show that every Hanner body has the stated volume product. The remaining work is the lower bound and its converse equality statement.

First let

$$A = \{X \in \mathbb{R}^d : |b_i \cdot X| \leq 1, 1 \leq i \leq m\}, \quad A^\circ = \text{conv}\{\pm b_1, \dots, \pm b_m\},$$

where the nonzero, pairwise nonproportional rows  $b_i$  span  $\mathbb{R}^d$ . Section 3 constructs a planar lens with horizontal half-width  $\lambda(t)$  at height  $t$ . For  $X \in \text{int } A$ , use this width to form

$$L_X = \{Y \in \mathbb{R}^d : |b_i \cdot Y| \leq \lambda(b_i \cdot X), 1 \leq i \leq m\}.$$

Each independent  $d$ -tuple of signed constraints that are tight at one point of  $L_X$  determines a simplex: take the convex hull of 0 and those signed rows. Let  $\Sigma_X$  be the union of these simplices. It lies in  $A^\circ$ .

Write  $F$  for the conformal map from the unit disk to the lens. For each set  $I$  of  $d$  independent rows, sample independent uniform angles  $\theta_i$  and solve  $b_i Z = F(e^{i\theta_i})$ ,  $i \in I$ , for the complex vector  $Z$ . Let  $P_I$  be the probability that all remaining row coordinates also lie in the closed lens. Section 5 applies the mass estimate of Section 4 to high powers of the inverse map: after a change of radial variables, the Jacobian integrals concentrate on these boundary samples and force  $S := \sum_I P_I \geq 1$ .

The uniform imaginary coordinate then identifies this probability sum with the simplex integral in Section 6:

$$1 \leq S = \frac{d!}{4^d} \int_A |\Sigma_X| dX \leq \frac{d!}{4^d} |A| |A^\circ|.$$

The first inequality is analytic; the last is the inclusion just described. The exact identity behind this bound also controls the missing volume  $|A^\circ| - |\Sigma_X|$ . Approximation extends the inequality to all origin-symmetric convex bodies.

For equality, set  $Q = K^\circ$  and pass to the body

$$B = K \oplus_1 [-1, 1], \quad B^\circ = Q \times [-1, 1]$$

in dimension  $d = n + 1$ . The norm-sum formula preserves equality at the new dimension. Inner polytope approximations to  $Q$  therefore have integrated missing volume tending to zero. Section 7 uses product neighborhoods to obtain, for each interior point of  $B$  and each nonzero direction, a representation whose rows attain their lens bounds at one vector satisfying all the constraints. This includes degenerate limiting representations. Finally, Section 8 examines points approaching an apex of  $B$ . The lens asymptotic gives a support-function inequality, and convex separation gives a metric median for every triple in the norm with unit ball  $Q$ : a point whose distances to each pair sum to that pair's distance. The Hansen–Lima theorem then identifies  $Q$ , and hence  $K$ , as a linear Hanner body [13, Corollary 7.4].

We next derive functional and entropy–transport consequences of the all-dimensional theorem. The geometric proof begins in Section 2 and ends in Section 8.

### 1.3 Functional Mahler inequality

The volume-product problem has a functional form in which polarity is replaced by convex conjugation. For a proper lower-semicontinuous convex function  $\varphi : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ , its Fenchel–Legendre transform is

$$\varphi^*(y) = \sup_{x \in \mathbb{R}^n} \{\langle x, y \rangle - \varphi(x)\}, \quad y \in \mathbb{R}^n.$$

We use the convention  $e^{-\infty} = 0$ .

**Corollary 1.2** (Even functional Mahler inequality). *For every integer  $n \geq 1$  and every even proper lower-semicontinuous convex function  $\varphi : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$  such that  $0 < \int_{\mathbb{R}^n} e^{-\varphi(x)} dx < \infty$ ,*

$$\left( \int_{\mathbb{R}^n} e^{-\varphi(x)} dx \right) \left( \int_{\mathbb{R}^n} e^{-\varphi^*(y)} dy \right) \geq 4^n.$$

*The second integral is allowed to be infinite, in which case the inequality is immediate.*

*Proof.* Set  $f = e^{-\varphi}$ , a nonzero integrable log-concave function. For  $z \in \mathbb{R}^n$ , Fradelizi and Meyer use the polar function

$$f^z(y) = \inf_{\{x: f(x) > 0\}} \frac{e^{-\langle x-z, y-z \rangle}}{f(x)}$$

and the translation-minimized functional volume product

$$\mathcal{P}(f) = \left( \int_{\mathbb{R}^n} f(x) dx \right) \inf_{z \in \mathbb{R}^n} \int_{\mathbb{R}^n} f^z(y) dy.$$

Their Proposition 1(a) states that the symmetric geometric Mahler inequality in every dimension implies  $\mathcal{P}(f) \geq 4^n$  for every even log-concave function on  $\mathbb{R}^n$  with  $0 < \int_{\mathbb{R}^n} f < \infty$  [7]. Theorem 1.1 supplies this all-dimensional hypothesis. Since  $f^0 = e^{-\varphi^*}$ , the product in Corollary 1.2 is at least  $\mathcal{P}(f)$ .  $\square$

The cited implication uses auxiliary convex bodies in dimensions  $n + m$  and lets  $m$  tend to infinity; it is not an application only of the geometric theorem in dimension  $n$ . The constant is sharp: for  $\varphi(x) = \sum_{i=1}^n |x_i|$ , the transform  $\varphi^*$  is zero on  $[-1, 1]^n$  and  $+\infty$  outside, and both integrals in the corollary equal  $2^n$ . The equality classification in Theorem 1.1 concerns convex bodies. The limiting implication does not by itself classify the even potentials for which equality holds [7, p. 1437].

For potentials without evenness, the companion paper on general convex bodies gives the sharp lower bound  $e^n$  [26, Corollary 1.2].

The functional inequality also has an entropy–transport formulation. For a log-concave probability measure  $\eta$  with a density, let  $H(\eta) = \int \log(d\eta/dx) d\eta$  denote its entropy relative to Lebesgue measure, the negative of differential Shannon entropy. For probability measures  $\mu_1, \mu_2$  with finite first moments, set

$$\mathcal{T}(\mu_1, \mu_2) = \inf_{f \in \mathcal{F}} \left\{ \int f d\mu_1 + \int f^* d\mu_2 \right\},$$

where  $\mathcal{F}$  is the class of proper lower-semicontinuous convex functions  $\mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$  and  $f^*$  is the Fenchel–Legendre transform. The cost  $\mathcal{T}$  is allowed to be  $+\infty$ .

**Corollary 1.3** (Symmetric entropy–transport inequality). *For every integer  $n \geq 1$ , let  $\eta_i(dx) = e^{-V_i(x)} dx$ ,  $i = 1, 2$ , be full-dimensional log-concave probability measures, both symmetric about the origin, with proper lower-semicontinuous convex potentials  $V_i$ . Suppose their densities are essentially continuous, meaning that  $e^{-V_i} = 0$  at  $\mathcal{H}^{n-1}$ -almost every point of  $\partial \text{supp } \eta_i$ . Their moment measures  $\nu_i = (\nabla V_i)_\# \eta_i$  have finite first moments,  $H(\eta_i) = -\int V_i d\eta_i$  is finite, and*

$$H(\eta_1) + H(\eta_2) \leq -n \log(4e^2) + \mathcal{T}(\nu_1, \nu_2).$$

*Proof.* For  $\rho_i = e^{-V_i}$ , the identity  $|\nabla \rho_i| = |\nabla V_i| e^{-V_i}$  holds almost everywhere on the interior of the support. Lemma 4 and the proof of Proposition 7 of [5] give  $\int |\nabla \rho_i| < \infty$  and  $V_i \in L^1(\eta_i)$ , respectively. The pushforward definition of  $\nu_i$  therefore gives its finite first moment, and  $H(\eta_i) = -\int V_i d\eta_i$  is finite. Gozlan’s equivalence between the symmetric functional inverse Santaló and entropy–transport inequalities, as stated in [8, Theorem 1.3], applies to Corollary 1.2 with  $c = 4$ .  $\square$

## 2 Polarity, norm sums, and Hanner polytopes

All bodies below are symmetric about the origin. In addition to the polar and volume product defined in Section 1, we use the gauge and support function

$$p_A(x) = \inf\{t > 0 : x \in tA\}, \quad h_A(y) = \max_{x \in A} \langle x, y \rangle.$$

Dual spaces are identified with Euclidean coordinate spaces, and  $|A|$  always denotes volume in the dimension of  $A$ . The notation  $h_A$  also applies to any nonempty compact convex set. The gauge is a norm with closed unit ball  $A$ . Convex separation gives the bipolar identity  $A^{\circ\circ} = A$ , and hence

$$p_A = h_{A^\circ}, \quad h_A = p_{A^\circ}. \tag{2}$$

Indeed  $x \in tA$  is equivalent to  $\langle x, y \rangle \leq t$  for every  $y \in A^\circ$ . For an invertible linear map  $T$ ,

$$(TA)^\circ = T^{-\top} A^\circ, \quad P(TA) = P(A), \tag{3}$$

since the two volumes acquire reciprocal determinant factors.

For bodies  $A \subset \mathbb{R}^k$  and  $A' \subset \mathbb{R}^l$ , define

$$A \oplus_\infty A' = A \times A', \quad A \oplus_1 A' = \text{conv}((A \times \{0\}) \cup (\{0\} \times A')).$$

These operations are taken in the product space. More generally, complementary subspaces are identified with product coordinates by an invertible linear map. Their gauges are

$$\begin{aligned} p_{A \oplus_\infty A'}(x, x') &= \max\{p_A(x), p_{A'}(x')\}, \\ p_{A \oplus_1 A'}(x, x') &= p_A(x) + p_{A'}(x'). \end{aligned} \quad (4)$$

For the second identity, a convex combination from the two coordinate bodies has gauge sum at most one. Conversely, when the sum is at most one, use these two gauges as the coefficients of the normalized coordinate vectors and place any remaining weight at 0. A zero coordinate contributes no term.

**Lemma 2.1** (Polars and volumes of the two sums). *For origin-symmetric convex bodies of dimensions  $k, l \geq 1$ ,*

$$(A \oplus_1 A')^\circ = A^\circ \oplus_\infty (A')^\circ, \quad (A \oplus_\infty A')^\circ = A^\circ \oplus_1 (A')^\circ. \quad (5)$$

Moreover,

$$|A \oplus_\infty A'| = |A| |A'|, \quad |A \oplus_1 A'| = \frac{k! l!}{(k+l)!} |A| |A'|. \quad (6)$$

Consequently, for  $s \in \{1, \infty\}$ ,

$$P(A \oplus_s A') = \frac{P(A)P(A')}{\binom{k+l}{s}}. \quad (7)$$

*Proof.* Testing a functional on the two coordinate bodies gives the first polar identity; taking polars again gives the second. The product volume follows from Fubini. By (4), the fiber of  $A \oplus_1 A'$  over  $x \in A$  is  $(1 - p_A(x))A'$ . Since  $|\{x : p_A(x) \leq t\}| = t^k |A|$  for  $0 \leq t \leq 1$ , integration gives

$$|A \oplus_1 A'| = k |A| |A'| \int_0^1 t^{k-1} (1-t)^l dt = \frac{k! l!}{(k+l)!} |A| |A'|.$$

The last integral is evaluated by repeated integration by parts. Combining the polar and volume identities proves (7).  $\square$

The Hanner recursion from Section 1 uses precisely these two operations, following Hanner [12]. Their volume and polarity formulas give the attaining family immediately.

**Lemma 2.2** (The Hanner volume product). *Every  $d$ -dimensional linear Hanner body  $H$  satisfies*

$$P(H) = \frac{4^d}{d!}.$$

*The polar of a linear Hanner body is again a linear Hanner body.*

*Proof.* A centered interval  $[-a, a]$ ,  $a > 0$ , has polar  $[-1/a, 1/a]$  and volume product 4. Equation (7) propagates  $4^k/k!$  and  $4^l/l!$  to  $4^{k+l}/(k+l)!$  under either operation. Induction and linear invariance prove the first assertion. The polar identities exchange the two operations and preserve the interval base case. Equation (3) then proves closure under polarity also for linear images.  $\square$

In dimension one every origin-symmetric convex body is a centered nondegenerate interval. Thus both the sharp inequality and the full equality classification in that dimension follow directly from the base case above.

### 3 A conformal lens with a uniform boundary coordinate

We construct a conformal image of the disk whose boundary has a uniformly distributed imaginary coordinate. This law will turn complex feasibility probabilities into real volumes. The horizontal half-width has further roles: strict curvature will make additional boundary equalities have probability zero, while concavity will extend feasibility from polar vertices to their convex hull. Its endpoint scaling will enter the equality argument near the apex of the lifted body. The map is a rotation of the uniform-distribution example in Gross's conformal Skorokhod embedding [11, Section 3.2]; we prove all its required properties directly.

Write  $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$  and define

$$F(z) = \frac{8}{\pi^2} \sum_{j=0}^{\infty} \frac{(-1)^j z^{2j+1}}{(2j+1)^2}. \quad (8)$$

The series converges uniformly on  $\overline{\mathbb{D}}$ .

**Lemma 3.1** (The planar lens). *The function  $F$  is a biholomorphism from  $\mathbb{D}$  onto a bounded convex domain  $D$ , with inverse  $g$ , and  $F'(0) = 8/\pi^2$ . There is a continuous even function  $\lambda : [-1, 1] \rightarrow [0, \infty)$ , positive on  $(-1, 1)$  and zero at  $\pm 1$ , such that*

$$\overline{D} = \{v + it : |t| \leq 1, |v| \leq \lambda(t)\}. \quad (9)$$

The function  $\lambda$  is smooth on  $(-1, 1)$  and satisfies

$$\lambda'(t) = -\frac{2}{\pi} \operatorname{arctanh}\left(\sin \frac{\pi t}{2}\right), \quad \lambda''(t) = -\sec \frac{\pi t}{2} < 0. \quad (10)$$

If  $\Theta$  is uniform on  $[0, 2\pi)$ , then

$$F(e^{i\Theta}) \stackrel{\text{law}}{=} \varepsilon \lambda(T) + iT, \quad (11)$$

where  $T$  is uniform on  $[-1, 1]$  and  $\varepsilon$  is an independent uniform sign.

*Proof.* The map is odd and commutes with conjugation. Put

$$w = \arctan z = \frac{1}{2i} \log \frac{1+iz}{1-iz}, \quad z \in \mathbb{D}.$$

The fraction inside the logarithm lies in the right half-plane, where we use the principal logarithm. Thus  $|\Re w| < \pi/4$ ,  $\tan w = z$ , and  $w = 0$  only when  $z = 0$ . Differentiating the series gives

$$F'(z) = \frac{8}{\pi^2} \frac{w}{z}, \quad 1 + \frac{zF''(z)}{F'(z)} = \frac{\sin(2w)}{2w}, \quad (12)$$

with their limiting values at zero. In particular  $F'$  has no zeros. For  $w = a + ib \neq 0$ , the real part of the last expression is

$$\frac{a \sin(2a) \cosh(2b) + b \cos(2a) \sinh(2b)}{2(a^2 + b^2)} > 0,$$

since  $|a| < \pi/4$ ; its value at zero is 1.

Fix  $0 < r < 1$  and put  $\gamma_r(\theta) = F(re^{i\theta})$ . This curve is regular, and its tangent angle  $\beta$  satisfies

$$\beta'(\theta) = \Re\left(1 + \frac{zF''(z)}{F'(z)}\right) > 0, \quad \beta(\theta + 2\pi) = \beta(\theta) + 2\pi.$$

The second identity uses the zero-free derivative in the disk. At a fixed parameter  $\theta_0$ , project the curve onto its right unit normal there. The derivative of this projection has the sign of  $-\sin(\beta(\theta) - \beta(\theta_0))$ . It is negative until the tangent has turned through  $\pi$ , and positive thereafter, until the curve returns to  $\theta_0$ . Hence the projection has its unique maximum at  $\theta_0$ . This proves simplicity and shows that every point of the curve is the unique supporting point of its convex hull in the corresponding normal direction. All normal directions occur, so the curve is the boundary of that convex hull and is positively oriented. The harmonic maximum principle for these supporting projections places  $F(r\mathbb{D})$  strictly inside the curve. The argument principle then gives exactly one preimage for every point inside and none outside. Thus  $F$  maps  $r\mathbb{D}$  biholomorphically onto its bounded convex interior  $D_r$ .

As  $r$  increases, these domains exhaust  $D = F(\mathbb{D})$ . Consequently  $F$  is univalent on  $\mathbb{D}$ ,  $D$  is convex, and its inverse  $g$  is holomorphic. For each  $0 < r < 1$ , continuity also gives  $\overline{D_r} = F(r\overline{\mathbb{D}}) \subset D$ . Continuity on the closed disk gives boundedness and  $\overline{D} = F(\overline{\mathbb{D}})$ . No boundary value  $F(\zeta)$ ,  $|\zeta| = 1$ , lies in  $D$ : otherwise continuity of  $g$  there and  $g(F(r\zeta)) = r\zeta$  would give  $g(F(\zeta)) = \zeta \notin \mathbb{D}$ . It follows that  $\partial D = F(\partial\mathbb{D})$ .

We now compute this boundary. On the right semicircle,  $z = e^{is}$  with  $-\pi/2 < s < \pi/2$ , one has

$$\frac{1 + iz}{1 - iz} = \frac{i \cos s}{1 + \sin s}, \quad \Re w = \frac{\pi}{4}, \quad \Im w = \frac{1}{2} \operatorname{arctanh}(\sin s).$$

The formulas for  $w$  and  $F'$  extend analytically across each compact subarc, so we may differentiate the boundary values. Writing  $F(e^{is}) = v(s) + it(s)$  gives

$$t'(s) = \frac{2}{\pi}, \quad v'(s) = -\frac{4}{\pi^2} \operatorname{arctanh}(\sin s).$$

Since  $t(0) = 0$ , we have  $t(s) = 2s/\pi$ . Define  $\lambda(t) = v(\pi t/2)$  and extend it to  $[-1, 1]$  by continuity. Conjugation makes  $\lambda$  even. The values  $F(\pm i)$  are purely imaginary by the series, so  $\lambda(\pm 1) = 0$ . Differentiation gives (10); strict concavity and the endpoint values then give positivity in the interior. Oddness and conjugation show that the other semicircle traces the graph  $v = -\lambda(t)$ . These two graphs give (9). On either semicircle the imaginary coordinate traverses  $[-1, 1]$  at constant absolute speed  $2/\pi$ . Each semicircle has probability  $1/2$ , which proves (11).  $\square$

Figure 1 illustrates the coordinate law.

The two boundary graphs are smooth away from  $\pm i$  and have strictly turning tangent directions by (10). In particular, a fixed tangent line direction occurs at only finitely many parameters away from these two endpoints. The boundary has planar Lebesgue measure zero, as is also immediate from its graph description.

**Lemma 3.2** (Endpoint scaling). *As  $\eta \downarrow 0$ ,*

$$\lambda(1 - \eta) = \frac{2}{\pi} \eta \log \frac{1}{\eta} + O(\eta). \quad (13)$$

*For every  $0 < a \leq b < \infty$ ,*

$$\sup_{a \leq c \leq b} \left| \frac{\lambda(1 - \delta c)}{\lambda(1 - \delta)} - c \right| \longrightarrow 0 \quad (\delta \downarrow 0). \quad (14)$$

*Proof.* Equation (10) and  $\operatorname{arctanh}(\cos x) = \log \cot(x/2)$  give

$$\lambda(1 - \eta) = \frac{2}{\pi} \int_0^\eta \log \cot \frac{\pi s}{4} ds.$$

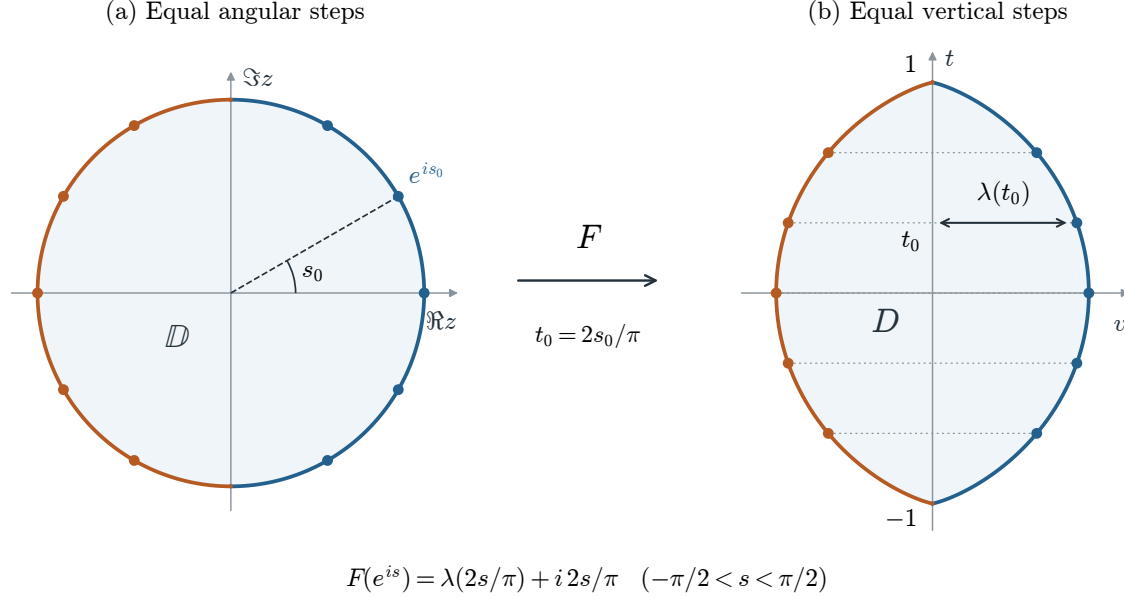


Figure 1: The conformal lens and its boundary coordinate. Equally spaced angles on either semicircle give equally spaced imaginary coordinates on the corresponding boundary branch. On the right semicircle,  $F(e^{is}) = \lambda(2s/\pi) + 2is/\pi$ ; reflection gives the left branch. Thus a uniform angle gives a uniform coordinate in  $[-1, 1]$  and an independent fair branch sign. The horizontal arrow marks the half-width  $\lambda(t_0)$ , with  $s_0 = \pi/6$  and  $t_0 = 1/3$ .

Here  $\log \cot(\pi s/4) = \log(1/s) + O(1)$  uniformly as  $s \downarrow 0$ . Integration proves (13). Writing  $L = \log(1/\delta)$ , the same uniform remainder gives

$$\frac{\lambda(1 - \delta c)}{\lambda(1 - \delta)} = c \frac{L - \log c + O(1)}{L + O(1)} = c + O_{a,b}(L^{-1})$$

uniformly for  $a \leq c \leq b$ , which proves (14).  $\square$

## 4 Mass at an isolated holomorphic zero

The next estimate turns the order of a holomorphic zero into an integral bound for its Jacobian minors. Its application will use high powers of the inverse planar map. In complex dimension  $d$ , the mass bound grows as the  $d$ th power of their exponent, matching the radial rescaling that will produce boundary feasibility probabilities in Section 5. The estimate is a special case of the sublevel-mass and comparison formulas for generalized Lelong numbers; see Demailly [6, Sections 3 and 5]. We give a direct Stokes proof in the normalization used here.

On  $\mathbb{C}^d$ , write  $z_j = x_j + iy_j$  and use the complex orientation

$$dV = dx_1 \wedge dy_1 \wedge \cdots \wedge dx_d \wedge dy_d.$$

For a smooth real function  $v$ , our convention is

$$d^c = \frac{i}{4}(\bar{\partial} - \partial), \quad dd^c = \frac{i}{2}\partial\bar{\partial}. \quad (15)$$

Thus  $dd^c v(\xi) = -\frac{1}{4}dv(i\xi)$  for a real tangent vector  $\xi$ , and

$$dd^c|z|^2 = \sum_{j=1}^d dx_j \wedge dy_j, \quad (dd^c v)^d = d! \det(v_{\bar{z}_i z_j}) dV. \quad (16)$$

The second identity follows by expanding the exterior product, or by a unitary diagonalization of the Hermitian Hessian.

**Lemma 4.1** (Holomorphic mass). *Let  $d \geq 1$ ,  $N \geq d$ , and let  $\mathcal{U} \subset \mathbb{C}^d$  be open and contain 0. Suppose that  $f = (f_1, \dots, f_N) : \mathcal{U} \rightarrow \mathbb{C}^N$  is holomorphic and has the following properties:*

1.  $f^{-1}(0) = \{0\}$ ;

2. for an integer  $k \geq 1$ ,

$$f(z) = H(z) + O(|z|^{k+1}) \quad \text{near } 0,$$

where each component of  $H$  is a homogeneous polynomial of degree  $k$  and  $H(z) \neq 0$  whenever  $z \neq 0$ ;

3. for every  $0 < s < 1$ , the set  $\{z \in \mathcal{U} : |f(z)|^2 \leq s\}$  is compact in  $\mathcal{U}$ .

Then, with  $\tau = |f|^2$ ,

$$\int_{\{\tau < 1\}} \sum_{\substack{I \subset \{1, \dots, N\} \\ |I|=d}} \left| \det \left( \frac{\partial f_i}{\partial z_j} \right)_{i \in I, 1 \leq j \leq d} \right|^2 dV \geq \frac{(\pi k)^d}{d!}. \quad (17)$$

All norms are Euclidean. Rows in each determinant are in increasing index order.

*Proof.* We first bound the integral of  $(dd^c \tau)^d$ . The conversion to Jacobian minors will then follow from Cauchy–Binet. Both  $\tau$  and  $\log \tau$  away from 0 have positive semidefinite complex Hessians. Indeed, if  $Z$  is the complex derivative matrix of  $f$ , the Hessian of  $\tau$  is  $Z^* Z$ . For a complex direction  $\xi$ , the Hessian of  $\log \tau$  evaluates to

$$\frac{|f|^2 |df(\xi)|^2 - |\langle df(\xi), f \rangle|^2}{|f|^4} \geq 0$$

by Cauchy–Schwarz. Their top exterior powers are therefore nonnegative.

**Comparison with a small sphere.** Fix a regular value  $s \in (0, 1)$  of  $\tau$  and set  $E_s = \{z \in \mathcal{U} : \tau(z) < s\}$ . By compactness of the sublevel and the regular-value theorem,  $\overline{E_s}$  is a compact smooth manifold with boundary  $\{\tau = s\}$ . Stokes’ theorem gives

$$\int_{E_s} (dd^c \tau)^d = \int_{\partial E_s} d^c \tau \wedge (dd^c \tau)^{d-1}.$$

Put  $\gamma = d^c \log \tau$  on  $\mathcal{U} \setminus \{0\}$ . The identities

$$d^c \log \tau = \tau^{-1} d^c \tau, \quad dd^c \log \tau = \tau^{-1} dd^c \tau - \tau^{-2} d\tau \wedge d^c \tau$$

and the vanishing of the pullback of  $d\tau$  to  $\partial E_s$  imply

$$s^{-d} \int_{E_s} (dd^c \tau)^d = \int_{\partial E_s} \gamma \wedge (d\gamma)^{d-1}. \quad (18)$$

For sufficiently small  $\varepsilon > 0$ , the closed Euclidean ball  $\overline{B_\varepsilon}$  is contained in  $E_s$ . Orient  $S_\varepsilon = \partial B_\varepsilon$  outward from this ball. Applying Stokes on  $E_s \setminus \overline{B_\varepsilon}$  gives

$$s^{-d} \int_{E_s} (dd^c \tau)^d - \int_{S_\varepsilon} \gamma \wedge (d\gamma)^{d-1} = \int_{E_s \setminus \overline{B_\varepsilon}} (dd^c \log \tau)^d \geq 0. \quad (19)$$

Thus it remains to determine the flux through a shrinking sphere.

**The homogeneous flux.** Let  $T(z) = |H(z)|^2$  and  $S = \{z : |z| = 1\}$ . The holomorphic Taylor expansion implies

$$\varepsilon^{-2k} \tau(\varepsilon z) \longrightarrow T(z) \quad \text{in } C^2 \text{ on a fixed closed annulus containing } S.$$

For example, termwise differentiation of the convergent Taylor series gives this convergence for the rescaled map and then for its squared norm. Since  $T$  is strictly positive on that annulus, the logarithms converge in  $C^2$  as well. Pulling back to  $S$  by dilation, the additive constant  $2k \log \varepsilon$  disappears under  $d^c$ . Consequently

$$\lim_{\varepsilon \downarrow 0} \int_{S_\varepsilon} \gamma \wedge (d\gamma)^{d-1} = \int_S \alpha \wedge (d\alpha)^{d-1}, \quad \alpha = d^c \log T|_S. \quad (20)$$

We compare this flux with that of the radial function  $|z|^{2k}$ . Homogeneity gives

$$T(rz) = r^{2k} T(z) \quad (r > 0), \quad T(e^{i\theta} z) = T(z).$$

Thus  $\log T - k \log |z|^2$  is unchanged by dilation and by common phase rotation. These invariances will make its contribution to the flux vanish. On  $S$ , put

$$\alpha_0 = d^c \log |z|^2|_S, \quad \beta = d^c (\log T - k \log |z|^2)|_S, \quad \alpha_t = k\alpha_0 + t\beta \quad (0 \leq t \leq 1).$$

Thus  $\alpha_1 = \alpha$ . For the nowhere-zero tangent vector field  $V(z) = iz$  on  $S$ , the dilation identity and (15) give  $\beta(V) = 0$  and  $\alpha_0(V) = 1/2$ . The phase-rotation identity makes every  $\alpha_t$  invariant under the flow of  $V$ . Cartan's identity therefore gives

$$\iota_V d\alpha_t = \mathcal{L}_V \alpha_t - d(\alpha_t(V)) = 0,$$

where  $\iota_V$  denotes contraction and  $\mathcal{L}_V$  the Lie derivative. It follows that the top-degree form  $\beta \wedge (d\alpha_t)^{d-1}$  on the  $(2d-1)$ -dimensional sphere vanishes: its contraction with the nonzero vector  $V$  is zero. Differentiation and integration by parts on the closed sphere now yield

$$\frac{d}{dt} \int_S \alpha_t \wedge (d\alpha_t)^{d-1} = d \int_S \beta \wedge (d\alpha_t)^{d-1} = 0.$$

For  $d = 1$  this is simply the derivative of  $\int_S \alpha_t$ ; the same argument applies.

On the unit sphere,  $\alpha_0$  and  $d\alpha_0$  are the restrictions of  $d^c |z|^2$  and  $dd^c |z|^2$ , respectively. Hence a final use of Stokes and (16) gives

$$\begin{aligned} \int_S \alpha \wedge (d\alpha)^{d-1} &= k^d \int_S \alpha_0 \wedge (d\alpha_0)^{d-1} \\ &= k^d \int_{\{|z| < 1\}} (dd^c |z|^2)^d \\ &= k^d d! \operatorname{vol}_{2d}(\{|z| < 1\}) = (\pi k)^d. \end{aligned}$$

**The minor bound.** Equations (19) and (20) imply

$$\int_{E_s} (dd^c \tau)^d \geq s^d (\pi k)^d$$

for every regular  $s \in (0, 1)$ . Sard's theorem supplies an increasing sequence of such values tending to 1. Since the integrand is nonnegative, monotone convergence gives

$$\int_{\{\tau < 1\}} (dd^c \tau)^d \geq (\pi k)^d.$$

Finally, the complex Hessian  $Z^*Z$  and Cauchy–Binet give the identity

$$(dd^c \tau)^d = d! \sum_{|I|=d} |\det Z_I|^2 dV.$$

Division by  $d!$  proves (17). □

## 5 Feasible bases

We now apply the holomorphic mass estimate to finitely many real strip constraints. It gives a lower bound for the sum of their boundary feasibility probabilities. For the lens of Lemma 3.1, these probabilities will become volumes of real simplices.

Fix  $d \geq 1$  and nonzero real row vectors  $b_1, \dots, b_m$  spanning  $(\mathbb{R}^d)^*$ . Write

$$A = \{X \in \mathbb{R}^d : |b_i X| \leq 1 \text{ for every } i\}, \quad C = A^\circ = \text{conv}\{\pm b_1, \dots, \pm b_m\}. \quad (21)$$

We identify rows with vectors when taking a convex hull, and extend their action complex-linearly to  $\mathbb{C}^d$ . The spanning assumption makes  $A$  bounded, and  $0 \in \text{int } A$ . The polar identity follows from the bipolar theorem: the polar of the displayed convex hull is exactly  $A$ . Let  $\mathcal{B}$  be the collection of  $d$ -element sets of independent rows. For  $I \in \mathcal{B}$ , put  $B_I = (b_i)_{i \in I}$  in increasing index order and  $D_I = |\det B_I|$ . Dependent row sets are omitted throughout.

For the next lemma,  $F$  may be any biholomorphism of the unit disk onto a bounded planar domain  $D$ , with  $F(0) = 0$  and a continuous extension to the closed disk. For independent uniform angles  $\theta_i \in [0, 2\pi)$ ,  $i \in I$ , define

$$Z_I(\theta) = B_I^{-1}(F(e^{i\theta_i}))_{i \in I}, \quad P_I = \mathbb{P}\{b_j Z_I(\theta) \in \overline{D} \text{ for all } j\}, \quad S = \sum_{I \in \mathcal{B}} P_I. \quad (22)$$

The active coordinates  $i \in I$  already belong to  $\overline{D}$ ; thus the probability tests only the remaining rows.

**Lemma 5.1** (Boundary feasibility). *Let  $d \geq 1$  and let  $b_1, \dots, b_m$  be nonzero real rows spanning  $(\mathbb{R}^d)^*$ . Let  $F : \mathbb{D} \rightarrow D$  be a biholomorphism onto a bounded planar domain, continuous on  $\overline{\mathbb{D}}$ , with  $F(0) = 0$ . Then the sum  $S$  in (22) satisfies  $S \geq 1$ . No condition on extra-row boundary equalities or on dependent row subsets is required.*

*Proof.* Let  $g = F^{-1}$  and set

$$\Omega = \{z \in \mathbb{C}^d : b_j z \in D \text{ for every } j\}, \quad h_j(z) = g(b_j z).$$

The set  $\Omega$  is open, bounded, and contains 0. For an integer  $k \geq 1$ , apply Lemma 4.1 to the holomorphic map

$$f_k = (h_1^k, \dots, h_m^k), \quad \tau_k = \sum_{j=1}^m |h_j|^{2k}.$$

We shall let  $k$  tend to infinity. The mass lower bound has scale  $k^d$ . The radial substitution below removes the same factor from the Jacobian integrals and leaves a fixed domain on which the inferred points tend to boundary samples. The unique zero of  $f_k$  is 0:  $g$  vanishes only at 0, and the rows span. Writing  $c = g'(0) \neq 0$ , its degree- $k$  leading term is  $((cb_1z)^k, \dots, (cb_mz)^k)$ , also nonzero whenever  $z \neq 0$ . For  $0 < s < 1$ , the set  $\{\tau_k \leq s\}$  is compact in  $\Omega$ . Indeed each of its row coordinates belongs to the fixed compact set  $F(\{|w| \leq s^{1/(2k)}\}) \subset D$ . A basis of rows bounds  $z$ ; limits retain all these inclusions and the inequality  $\tau_k \leq s$ . All hypotheses of the mass lemma therefore hold. The derivative row of  $h_j^k$  is a scalar multiple of  $b_j$ , so dependent row subsets have zero minors. Consequently

$$\sum_{I \in \mathcal{B}} M_{I,k} \geq k^d \frac{\pi^d}{d!}, \quad M_{I,k} := \int_{\{\tau_k < 1\}} \left| \det \frac{\partial (h_i^k)_{i \in I}}{\partial (z_1, \dots, z_d)} \right|^2 dV_{2d}(z). \quad (23)$$

Fix  $I$ . The change of variables  $u_i = h_i(z)$ ,  $i \in I$ , is one-to-one onto its image, with inverse  $z = B_I^{-1}(F(u_i))_{i \in I}$ . Its squared complex Jacobian is its real volume Jacobian. The squared determinant in  $M_{I,k}$  is this Jacobian factor times  $k^{2d} \prod_i |u_i|^{2k-2}$ . The transformed integration domain is contained in

$$\left\{ u \in \mathbb{D}^d : \sum_{i \in I} |u_i|^{2k} < 1, \quad b_j B_I^{-1}(F(u_i))_{i \in I} \in D \text{ for all } j \right\}.$$

We use this larger domain for an upper bound, and then put  $u_i = \rho_i^{1/k} e^{i\theta_i}$ . The coordinate hyperplanes, on which polar coordinates are undefined, have Lebesgue measure zero. Since

$$k^2 |u_i|^{2k-2} dV_2(u_i) = k \rho_i d\rho_i d\theta_i,$$

we obtain

$$\frac{M_{I,k}}{k^d} \leq \int_{\rho_i > 0, \sum_i \rho_i^2 < 1} \mathbf{1}_{\left\{ b_j B_I^{-1}(F(\rho_i^{1/k} e^{i\theta_i}))_{i \in I} \in D \text{ for every } j \right\}} \prod_{i \in I} \rho_i d\rho_i d\theta_i. \quad (24)$$

The measure in this integral is finite: its total mass is the volume  $\pi^d/d!$  of the unit ball in  $\mathbb{C}^d$ . For every positive  $\rho_i$ , continuity of  $F$  on the closed disk makes the inferred points converge to  $Z_I(\theta)$ . If a limiting row coordinate lies outside  $\overline{D}$ , that constraint fails for all sufficiently large  $k$ . Thus the upper limit of the indicator is bounded by the indicator of the event defining  $P_I$ , including all possible boundary equalities. The upper-bound form of Fatou's lemma, applied to functions bounded by 1 on this finite measure space, gives

$$\limsup_{k \rightarrow \infty} \frac{M_{I,k}}{k^d} \leq \frac{\pi^d}{d!} P_I.$$

There are only finitely many bases, so (23) and these upper bounds give

$$\begin{aligned} \frac{\pi^d}{d!} &\leq \limsup_{k \rightarrow \infty} \sum_{I \in \mathcal{B}} \frac{M_{I,k}}{k^d} \\ &\leq \sum_{I \in \mathcal{B}} \limsup_{k \rightarrow \infty} \frac{M_{I,k}}{k^d} \\ &\leq \frac{\pi^d}{d!} \sum_{I \in \mathcal{B}} P_I. \end{aligned}$$

Thus  $S \geq 1$ . □

### 5.1 Boundary equalities for the lens

The lower bound for  $S$  allows additional boundary equalities. The real volume identity in Section 6 will need the stronger fact that these equalities have probability zero. From now on,  $F, D$ , and  $\lambda$  are those of Lemma 3.1. We may also require that no two rows are proportional. For any finite strip presentation this is achieved without changing  $A$  by keeping, on each row line, a row of largest magnitude; its strip implies all the discarded strips on that line. Zero rows, if present initially, impose no constraint and are discarded. In dimension one the reduced list has a single row, so there are no extra-row boundary equalities to consider.

**Lemma 5.2** (Additional boundary equalities have probability zero). *Let  $F : \mathbb{D} \rightarrow D$  be the lens map of Lemma 3.1. Assume that the nonzero spanning rows  $b_1, \dots, b_m$  are pairwise nonproportional. For every  $I \in \mathcal{B}$  and  $j \notin I$ ,*

$$\mathbb{P}\{b_j Z_I(\theta) \in \partial D\} = 0.$$

*Proof.* Write  $b_j B_I^{-1} = (c_i)_{i \in I}$ . At least two coefficients, say  $c_a, c_b$ , are nonzero; otherwise  $b_j$  would be proportional to a basis row. Let  $\gamma(\theta) = F(e^{i\theta})$ . Apart from its two endpoints, the lens boundary consists of the smooth regular branches  $(\pm \lambda(t), t)$ ,  $-1 < t < 1$ . Their tangent directions have at most one occurrence on each branch for any fixed unoriented line direction, because  $\lambda'$  is strictly monotone. The boundary parameter has constant nonzero speed in  $t$  on each branch, as established in the proof of Lemma 3.1.

Hold all phases other than  $\theta_a, \theta_b$  fixed. The map

$$(\theta_a, \theta_b) \mapsto c_a \gamma(\theta_a) + c_b \gamma(\theta_b) + \sum_{i \neq a, b} c_i \gamma(\theta_i)$$

has real Jacobian determinant  $\det(c_a \gamma'(\theta_a), c_b \gamma'(\theta_b))$ . For every nonendpoint  $\theta_a$ , it vanishes at only finitely many nonendpoint values of  $\theta_b$ . Its zero set, including the excluded endpoint lines, therefore has planar measure zero by Fubini's theorem. Off that set the inverse function theorem gives local diffeomorphism charts. A countable subcover of such charts suffices, and in each chart the inverse is locally Lipschitz, so it sends a planar null set to a null set. The set  $\partial D$  is planar-null, being a countable union of compact smooth subarcs and two points. Its inverse image consequently has measure zero in the two remaining phases. A final application of Fubini over the fixed phases proves the assertion. □

## 6 The real simplex identity

The lens boundary law now turns the feasibility sum into an exact integral over the strip body. The simplices in this integral lie in the polar body; their missing volume will also be the quantity used in the equality argument.

Keep the nonzero spanning, pairwise nonproportional rows of (21). For  $X \in \text{int } A$ , define

$$L_X = \{Y \in \mathbb{R}^d : |b_j Y| \leq \lambda(b_j X) \text{ for every } j\}. \quad (25)$$

All widths are strictly positive, so  $L_X$  contains a neighborhood of 0; a basis of rows shows that it is bounded. It is therefore a convex polytope with nonempty interior. For a basis  $I$  and signs  $\epsilon = (\epsilon_i)_{i \in I} \in \{-1, 1\}^I$ , let

$$Y_{I,\epsilon}(X) = B_I^{-1}(\epsilon_i \lambda(b_i X))_{i \in I}.$$

We call  $(I, \epsilon)$  feasible at  $X$  when this vector belongs to  $L_X$ . This definition specifies a unique vector for every choice, whether or not it is feasible. Set

$$T_{I,\epsilon} = \text{conv}(0, \{\epsilon_i b_i : i \in I\}), \quad \Sigma_X = \bigcup_{(I,\epsilon) \text{ feasible at } X} T_{I,\epsilon}. \quad (26)$$

Thus  $\Sigma_X$  is a finite union of closed  $d$ -simplices contained in  $C = A^\circ$ . This definition applies at every interior  $X$ , including points with additional tight constraints. For integration only, declare every pair infeasible and set  $\Sigma_X = \emptyset$  when  $X \in \partial A$ .

A feasible pair specifies a vertex  $Y_{I,\epsilon}(X)$  of  $L_X$ . A strictly positive combination of its signed tight rows defines a linear functional maximized uniquely at that vertex. This is why different feasible pairs give simplices with disjoint interiors whenever their vertices are distinct. The proof below removes, outside a null set of  $X$ , the boundary ties that could make two pairs specify the same vertex.

**Proposition 6.1** (Feasible-simplex identity). *Let  $b_1, \dots, b_m$  be nonzero, pairwise nonproportional real rows spanning  $(\mathbb{R}^d)^*$ , and let  $F, D, \lambda$  be as in Lemma 3.1. With  $A, S$ , and  $\Sigma_X$  defined in (21), (22), and (26), respectively,*

$$1 \leq S = \frac{d!}{4^d} \int_A |\Sigma_X| dX \leq \frac{d!}{4^d} |A| |A^\circ|. \quad (27)$$

In particular,  $|A| |A^\circ| \geq 4^d/d!$ , and

$$0 \leq \int_A (|A^\circ| - |\Sigma_X|) dX \leq |A| |A^\circ| - \frac{4^d}{d!}. \quad (28)$$

*Proof.* We first convert each probability into a real integral. By (11), each active boundary coordinate has the law  $\epsilon_i \lambda(t_i) + it_i$ , with  $t_i$  uniform on  $[-1, 1]$  and  $\epsilon_i$  an independent fair sign. Consequently each fixed sign choice has joint measure  $4^{-d} dt_I$  on  $[-1, 1]^d$ . Writing the inferred complex vector as  $Y + iX$  gives

$$t_I = B_I X, \quad Y = Y_{I,\epsilon}(X).$$

The constraints that all row coordinates lie in  $\overline{D}$  are exactly  $X \in A$  and  $Y \in L_X$ , apart from the immaterial boundary  $\partial A$ . This boundary has measure zero, since it is contained in the finitely many hyperplanes  $b_j X = \pm 1$ . The change of variables  $t_I = B_I X$  therefore gives

$$P_I = \frac{D_I}{4^d} \int_A \sum_{\epsilon \in \{-1, 1\}^I} \mathbf{1}\{(I, \epsilon) \text{ is feasible at } X\} dX. \quad (29)$$

For almost every  $X \in \text{int } A$ , every feasible pair  $(I, \epsilon)$  has no tight constraint outside  $I$ . Indeed, such an additional equality means that an extra row of  $Y + iX$  lies on  $\partial D$ . Lemma 5.2 makes this event null in the basis phases; the boundary law and the invertible change  $t_I = B_I X$  make its set of

$X$  null for each fixed sign choice. There are finitely many bases, sign choices, and extra rows, so a single null set removes all these equalities.

Fix  $X$  outside this null set. Distinct feasible pairs have distinct vectors  $Y_{I,\epsilon}(X)$ . If their bases differ, a shared vector would have a tight row outside one of the bases. If their bases agree but some signs differ, the shared vector would satisfy opposite equations with the same strictly positive width, which is impossible.

The interiors of their simplices are disjoint. To see this, let  $e \in \text{int } T_{I,\epsilon}$ . It has a representation  $e = \sum_{i \in I} \alpha_i \epsilon_i b_i$  with all  $\alpha_i > 0$  and  $\sum_i \alpha_i < 1$ . For every  $Y \in L_X$ ,

$$eY \leq \sum_{i \in I} \alpha_i \lambda(b_i X),$$

and equality holds at  $Y_{I,\epsilon}(X)$ . Equality forces every independent equation  $\epsilon_i b_i Y = \lambda(b_i X)$ , so this is the unique maximizer of the linear functional  $e$  on  $L_X$ . If  $e$  belonged to the interiors of two feasible simplices, their distinct vectors would both be its unique maximizer, a contradiction. Each simplex has volume  $D_I/d!$ , and their boundaries have measure zero. It follows that, for almost every  $X$ ,

$$d! |\Sigma_X| = \sum_{I \in \mathcal{B}} \sum_{\epsilon \in \{-1,1\}^I} D_I \mathbf{1}\{(I, \epsilon) \text{ is feasible at } X\}. \quad (30)$$

All integrals here are measurable. For each pair, feasibility is a finite collection of closed inequalities between continuous functions of  $X \in \text{int } A$ . In particular the set

$$\{(X, e) : X \in \text{int } A, e \in \Sigma_X\}$$

is a finite union of products of Borel feasibility sets and fixed closed simplices. Fubini's theorem makes  $X \mapsto |\Sigma_X|$  measurable, and it is bounded by  $|C|$ . Sum (29), use (30), and apply Lemma 5.1. This proves the first equality and lower bound in (27). The containment  $\Sigma_X \subset C$  proves the upper bound. In particular, the missing volume has the exact expression

$$\int_A (|C| - |\Sigma_X|) dX = |A| |C| - \frac{4^d}{d!} S.$$

It is nonnegative by containment, and  $S \geq 1$  bounds it above by  $|A| |C| - 4^d/d!$ . This proves (28) and identifies the quantity that will tend to zero in the equality argument.  $\square$

For a concrete union with positive missing volume, take  $b_1 = (1, 0)$ ,  $b_2 = (1/2, \sqrt{3}/2)$ ,  $b_3 = b_2 - b_1$ , and  $X = 0.9(1, 1/\sqrt{3})$ , as in Figure 2. Then  $b_1 X = b_2 X = 0.9$  and  $b_3 X = 0$ . Write  $a = \lambda(0.9)$ . The endpoint integral for  $\lambda$  and  $\cot x < 1/x$  for  $0 < x < \pi/2$  give

$$a < \frac{\log(40/\pi) + 1}{5\pi} < \frac{4}{15}, \quad \lambda(0) > \frac{8}{\pi^2} \left(1 - \frac{1}{9}\right) > \frac{32}{45}.$$

The second bound uses the alternating series for  $F(1)$ . In particular,  $2a < \lambda(0)$ , so the third strip in the example is strictly redundant.

## 6.1 Approximation of an arbitrary convex body

**Corollary 6.2** (The symmetric volume-product inequality). *For every origin-symmetric convex body  $K \subset \mathbb{R}^d$ ,  $d \geq 1$ ,*

$$|K| |K^\circ| \geq \frac{4^d}{d!}.$$

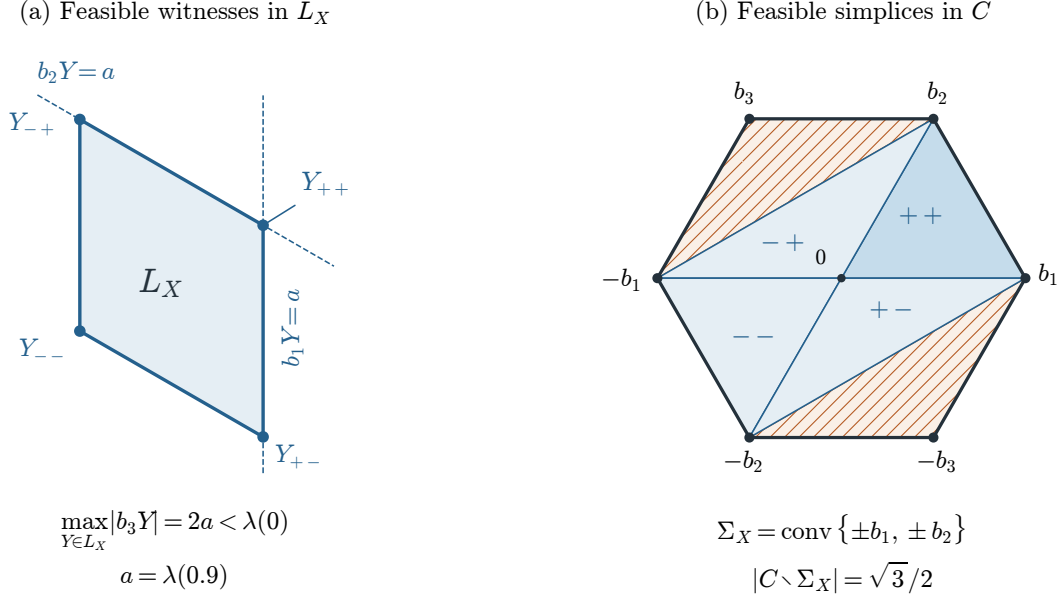


Figure 2: A feasible-simplex union can leave part of the polar uncovered. Here  $b_1 = (1, 0)$ ,  $b_2 = (1/2, \sqrt{3}/2)$ ,  $b_3 = (-1/2, \sqrt{3}/2)$ ,  $C = \text{conv}\{\pm b_1, \pm b_2, \pm b_3\}$ , and  $X = 0.9(1, 1/\sqrt{3})$ . The third strip is redundant in  $L_X$ , as verified above. The four witnesses  $Y_{\epsilon_1 \epsilon_2}$ , defined by  $b_i Y = \epsilon_i a$  for  $i = 1, 2$ , correspond to the four triangles  $\text{conv}(0, \epsilon_1 b_1, \epsilon_2 b_2)$  with matching sign labels. Their union is  $\Sigma_X = \text{conv}\{\pm b_1, \pm b_2\}$ . The two hatched caps have total area  $\sqrt{3}/2$ , one third of  $|C|$ .

*Proof.* Put  $Q = K^\circ$  and choose  $a > 0$  with  $aB_2^d \subset Q$ , where  $B_2^d$  is the Euclidean unit ball. For  $\varepsilon_j \downarrow 0$ , take a finite  $\varepsilon_j$ -net in  $Q$ , adjoin its negatives, and let  $Q_j$  be its convex hull. Thus  $Q_j \subset Q$ , and their support functions satisfy, for every Euclidean unit vector  $u$ ,

$$h_Q(u) - \varepsilon_j \leq h_{Q_j}(u) \leq h_Q(u), \quad h_Q(u) \geq a.$$

For  $\delta_j = \varepsilon_j/a < 1$ , this implies

$$(1 - \delta_j)Q \subset Q_j \subset Q, \quad K \subset A_j := Q_j^\circ \subset (1 - \delta_j)^{-1}K. \quad (31)$$

The first inclusions follow from the support-function inequalities and convex separation; the second follow by polarity. In particular  $Q_j$  and  $A_j$  are full-dimensional symmetric polytopes for large  $j$ .

Choose one vertex from each antipodal vertex pair of  $Q_j$  as a row list. These rows are nonzero and span. They are pairwise nonproportional: two distinct boundary vertices on the same ray from an interior origin cannot both be extreme, and the antipodal duplication has already been removed. Their strip body is  $A_j$ . Proposition 6.1 therefore gives  $|A_j| |Q_j| \geq 4^d/d!$ . The homothetic bounds (31) show both  $|Q_j| \rightarrow |Q|$  and  $|A_j| \rightarrow |K|$ , because dilation by  $t > 0$  multiplies volume by  $t^d$ . Passing to the limit proves the assertion. No smoothness, simplicity, or general-position assumption is imposed on  $K$ . In dimension one the assertion can also be read directly:  $K = [-a, a]$  has polar  $[-1/a, 1/a]$  and product 4.  $\square$

## 7 Equality and a common feasibility witness

The integral in (27) measures how much of the polar body is covered by feasible simplices. For a body attaining the sharp volume product, we shall make the uncovered volume tend to zero after

adjoining one segment. Compactness then gives a feasible witness at every interior point and in every direction. The resulting representation will be the input to the equality classification.

Throughout this section  $K = -K \subset \mathbb{R}^n$  is a convex body,  $n \geq 1$ , satisfying

$$|K| |K^\circ| = \frac{4^n}{n!}.$$

Set  $Q = K^\circ$ ,  $d = n + 1$ , and

$$B = K \oplus_1 [-1, 1] = \{(x, s) : p_K(x) + |s| \leq 1\}, \quad C = B^\circ = Q \times [-1, 1]. \quad (32)$$

The polar  $C$  has the two horizontal faces  $Q \times \{1\}$  and  $Q \times \{-1\}$ . The representations constructed below will use vectors from these faces. In this lifted space,  $X$  and  $Y$  belong to  $\mathbb{R}^d$ , while elements of  $C$  act on them by the Euclidean pairing. Recall that the lens width  $\lambda : [-1, 1] \rightarrow [0, \infty)$  is continuous, even and concave, with  $\lambda(0) > 0$  and  $\lambda(t) \leq \lambda(0)$ .

**Proposition 7.1** (A representation minimizing the lens cost). *Let  $K \subset \mathbb{R}^n$ ,  $n \geq 1$ , be an origin-symmetric convex body with  $P(K) = 4^n/n!$ . Put  $Q = K^\circ$ ,  $d = n + 1$ , and define the lifted bodies  $B, C$  by (32). For every  $X \in \text{int } B$  and every nonzero  $\xi = (q, a) \in \mathbb{R}^n \times \mathbb{R}$ , put*

$$T = p_C(\xi) = \max\{p_Q(q), |a|\}.$$

*There are  $1 \leq r \leq d$  vectors  $c_i = (b_i, \sigma_i)$  with  $b_i \in Q$  and  $\sigma_i \in \{-1, 1\}$ , nonnegative numbers  $\alpha_i$ , and a vector  $Y \in \mathbb{R}^d$ , such that*

$$\xi = \sum_{i=1}^r \alpha_i c_i, \quad \sum_{i=1}^r \alpha_i = T, \quad (33)$$

*and*

$$|h \cdot Y| \leq \lambda(h \cdot X) \quad (h \in C), \quad c_i \cdot Y = \lambda(c_i \cdot X) \quad (1 \leq i \leq r). \quad (34)$$

*In particular, for every finite representation  $\xi = \sum_{j=1}^s \beta_j h_j$  with  $\beta_j \geq 0$  and  $h_j \in C$ ,*

$$\sum_{i=1}^r \alpha_i \lambda(c_i \cdot X) \leq \sum_{j=1}^s \beta_j \lambda(h_j \cdot X). \quad (35)$$

*The choices may depend on  $X$  and  $\xi$ ; the same choice satisfies (35) for all the competing representations.*

The last coordinates  $\sigma_i$  split the chosen rows into two groups. In Section 8, we compare representation costs as  $X$  approaches the apex  $(0, 1)$  of  $B$ ; the endpoint behavior of  $\lambda$  makes the two signs contribute differently. We now prove Proposition 7.1 in three steps: approximate  $C$  by polytopes, locate feasible simplices near each prescribed point and direction, and pass to a common witness to compare costs.

## 7.1 Inner polytopes and vanishing missing volume

The horizontal section of  $B$  at height  $s \in [-1, 1]$  is  $(1 - |s|)K$ . Hence

$$|B| = |K| \int_{-1}^1 (1 - |s|)^n ds = \frac{2|K|}{n+1}, \quad |C| = 2|Q|,$$

and therefore

$$|B| |C| = \frac{4}{n+1} |K| |Q| = \frac{4^d}{d!}. \quad (36)$$

Choose full-dimensional symmetric polytopes  $Q_j \subset Q$  converging to  $Q$  in Hausdorff distance. For example, take the convex hulls of increasingly fine finite symmetric nets in  $Q$ , including a fixed spanning symmetric set. There are numbers  $t_j \in (0, 1)$  with  $t_j \rightarrow 1$  such that

$$t_j Q \subseteq Q_j \subseteq Q.$$

Indeed, if  $Q$  contains a Euclidean ball of radius  $r_0 > 0$  and the Hausdorff error is  $\varepsilon_j$ , then on Euclidean unit directions

$$h_{Q_j} \geq h_Q - \varepsilon_j \geq (1 - \varepsilon_j/r_0)h_Q.$$

The support-function criterion for containment gives the claim, after discarding finitely many indices and choosing  $t_j$  slightly smaller if necessary.

Put  $C_j = Q_j \times [-1, 1]$  and  $B_j = C_j^\circ$ . The inclusions above give

$$t_j C \subseteq C_j \subseteq C, \quad B \subseteq B_j \subseteq t_j^{-1} B. \quad (37)$$

In particular,  $|B_j|/|C_j| \rightarrow |B|/|C|$ . Use one row from each antipodal vertex pair of  $C_j$  to describe  $B_j$  as an intersection of strips. These rows span  $\mathbb{R}^d$ , and no two are proportional: a ray from the interior point 0 meets the boundary of a convex body in only one point. Every signed row is a vertex of  $C_j$ , and thus belongs to  $Q_j \times \{-1, 1\}$ .

Let  $\Sigma_{j,X}$  be the union of the feasible signed basis simplices for this strip description, as in (27). In particular, for every  $X \in \text{int } B_j$  it is a finite union of closed simplices contained in  $C_j$ . Its definition is retained also at the exceptional interior points where several constraints are simultaneously tight. The integral estimate and (36) imply

$$\begin{aligned} 0 \leq D_j &:= \int_{B_j} (|C_j| - |\Sigma_{j,X}|) dX \\ &\leq |B_j| |C_j| - \frac{4^d}{d!} \rightarrow 0. \end{aligned} \quad (38)$$

## 7.2 From an integral deficit to each point and direction

Fix  $X \in \text{int } B$  and  $e \in \partial C$ . We first construct indices  $j_k \rightarrow \infty$  and points

$$X_k \in \text{int } B_{j_k}, \quad e_k \in \Sigma_{j_k, X_k}, \quad (X_k, e_k) \rightarrow (X, e). \quad (39)$$

This is a consequence of (38) on product neighborhoods; it does not require a pointwise limit of the functions  $X \mapsto |\Sigma_{j,X}|$ .

For each  $k$ , choose a Euclidean ball  $U_k$  of positive volume contained in  $\text{int } B \cap B(X, 1/k)$ . There is also a ball  $V_k$  of positive volume with closure contained in  $\text{int } C \cap B(e, 1/k)$ . To see this at the boundary point  $e$ , first move a small distance from  $e$  toward 0; the resulting point is interior because  $0 \in \text{int } C$ . Take a sufficiently small ball about that point. By (37),  $U_k \subset \text{int } B_j$  for every  $j$ , and  $V_k \subset C_j$  for all sufficiently large  $j$ . The second assertion follows also from  $\max_{h \in \overline{V_k}} p_C(h) < 1$  and  $t_j \rightarrow 1$ .

Choose  $j_k > j_{k-1}$  so large that these inclusions hold and  $D_{j_k} < |U_k| |V_k|$ . If  $\Sigma_{j_k, Z}$  missed  $V_k$  for every  $Z \in U_k$ , its missing volume would be at least  $|V_k|$  throughout  $U_k$ , contradicting this inequality. Thus choose  $X_k \in U_k$  and  $e_k \in \Sigma_{j_k, X_k} \cap V_k$ . This proves (39).

Choose one feasible simplex containing  $e_k$ . Its  $d$  signed basis rows, written  $c_{1k}, \dots, c_{dk} \in Q_{j_k} \times \{-1, 1\}$ , give weights  $\theta_{ik} \geq 0$  satisfying

$$e_k = \sum_{i=1}^d \theta_{ik} c_{ik}, \quad \sum_{i=1}^d \theta_{ik} \leq 1. \quad (40)$$

The coefficient left over is the weight of the simplex vertex 0. Feasibility supplies one vector  $Y_k$  such that

$$c_{ik} \cdot Y_k = \lambda(c_{ik} \cdot X_k) \quad (1 \leq i \leq d), \quad |v \cdot Y_k| \leq \lambda(v \cdot X_k) \quad \text{for every vertex } v \text{ of } C_{j_k}.$$

Here a sign on a basis row has been absorbed using the evenness of  $\lambda$ .

The last inequality extends to all  $h \in C_{j_k}$ . Indeed, write  $h = \sum_{\ell} s_{\ell} v_{\ell}$  as a convex combination of vertices. The triangle inequality followed by concavity gives

$$|h \cdot Y_k| \leq \sum_{\ell} s_{\ell} |v_{\ell} \cdot Y_k| \leq \sum_{\ell} s_{\ell} \lambda(v_{\ell} \cdot X_k) \leq \lambda(h \cdot X_k).$$

Consequently  $Y_k \in \lambda(0)B_{j_k}$ . The polar sandwich (37) makes these witnesses uniformly bounded, even if the selected bases approach singularity.

Pass to a subsequence on which  $Y_k \rightarrow Y$ , every  $c_{ik} \rightarrow c_i$ , and every  $\theta_{ik} \rightarrow \theta_i$ . This uses a fixed finite product of compact sets: all rows belong to  $Q \times \{-1, 1\}$  and all weights belong to  $[0, 1]$ . No independence of the limiting rows is needed. We have

$$c_i \in Q \times \{-1, 1\}, \quad e = \sum_{i=1}^d \theta_i c_i, \quad \theta_i \geq 0, \quad \sum_i \theta_i \leq 1,$$

and continuity of  $\lambda$  gives  $c_i \cdot Y = \lambda(c_i \cdot X)$  for all  $i$ . For arbitrary  $h \in C$ , use  $h_k = t_{j_k} h \in C_{j_k}$  in the feasibility inequality and pass to the limit. It follows that

$$|h \cdot Y| \leq \lambda(h \cdot X) \quad (h \in C). \quad (41)$$

The subsequence has already been fixed; this argument proves (41) for all  $h$  with the same  $Y$ .

Since  $e \in \partial C$ , its gauge is one. Therefore

$$1 = p_C(e) \leq \sum_{i=1}^d \theta_i p_C(c_i) \leq \sum_{i=1}^d \theta_i \leq 1.$$

Thus  $\sum_i \theta_i = 1$ : none of the limiting mass remains at the simplex vertex 0.

### 7.3 Comparison with every positive representation

To prove Proposition 7.1, fix its  $X$  and nonzero  $\xi$ , and apply the preceding construction to  $e = \xi/T \in \partial C$ . Set  $\alpha_i = T\theta_i$  and discard any zero weights if desired. This gives at most  $d$  rows and proves (33) and (34). For any finite competing representation  $\xi = \sum_j \beta_j h_j$ ,  $\beta_j \geq 0$ ,  $h_j \in C$ , the common witness gives

$$\sum_i \alpha_i \lambda(c_i \cdot X) = \xi \cdot Y = \sum_j \beta_j h_j \cdot Y \leq \sum_j \beta_j \lambda(h_j \cdot X).$$

This proves (35) and completes the proof of Proposition 7.1. In particular, the conclusion holds for each prescribed  $X$  and  $\xi$ , including directions for which every limiting representation is degenerate.

## 8 From endpoint costs to metric medians

We now convert the optimal representations into a geometric property of the polar body. A *metric median* of three points in a normed space is a point that lies between every pair: if the points are  $x_0, x_1, x_2$ , a median  $m$  satisfies

$$\|x_i - x_j\| = \|x_i - m\| + \|m - x_j\| \quad (0 \leq i < j \leq 2).$$

The median need not be unique. Its existence will imply the ball intersection property that characterizes linear Hanner bodies.

**Proposition 8.1** (Medians of an equality body). *Let  $K \subset \mathbb{R}^n$ ,  $n \geq 1$ , be an origin-symmetric convex body with  $P(K) = 4^n/n!$ , and put  $Q = K^\circ$ . Every triple of points in  $(\mathbb{R}^n, p_Q)$  has a metric median.*

*Proof.* Write  $\|\cdot\| = p_Q$ . Translation reduces the triple to  $0, x, y$ . If two points coincide, that repeated point is a median, so assume that  $0, x, y$  are distinct. Set

$$p_1 = \|x\|, \quad p_2 = \|y\|, \quad p = \|x - y\|, \quad s = p_1 + p_2, \quad a = p_1 - p_2.$$

All three distances are positive. The triangle and reverse triangle inequalities give  $s \geq p \geq |a|$ . Define

$$A_+ = \frac{p+a}{2}, \quad A_- = \frac{p-a}{2}, \quad c = \frac{s-p}{2}. \quad (42)$$

These numbers are nonnegative, with  $A_+ + A_- = p$ ,  $c + A_+ = p_1$ , and  $c + A_- = p_2$ .

**The compact set of aggregate directions.** Let

$$D_0 = \{r_+ - r_- : r_+ + r_- = x - y, \quad \|r_+\| \leq A_+, \quad \|r_-\| \leq A_-\}. \quad (43)$$

Our goal is to prove  $x + y \in D_0 + (s - p)Q$ . Such membership supplies  $r_+, r_-$  for which  $m = x - r_+ = y + r_-$  has distances to  $0, x, y$  bounded by  $c, A_+, A_-$ , respectively; the triangle inequalities then force the three median equalities. We shall prove this membership by estimating support functions in each direction.

The set  $D_0$  is compact and convex: it is the linear image of the intersection of  $A_+Q \times A_-Q$  with the affine subspace  $r_+ + r_- = x - y$ . It is nonempty, since  $r_\pm = (A_\pm/p)(x - y)$  satisfy the constraints. This also covers  $A_+ = 0$  or  $A_- = 0$ .

**Endpoint comparison.** Fix  $u \in \text{int } K$  and let  $X_\delta = (\delta u, 1 - \delta)$ ,  $0 < \delta < 1$ . The lifted body  $B = K \oplus_1 [-1, 1]$  contains  $X_\delta$  in its interior because

$$p_B(X_\delta) = \delta p_K(u) + 1 - \delta < 1.$$

Apply Proposition 7.1 with this  $X_\delta$  and  $(q, a) = (x - y, p_1 - p_2)$ . Its total weight is  $T = \max\{\|q\|, |a|\} = p$ . We obtain rows  $(b_i, \sigma_i) \in Q \times \{-1, 1\}$  and weights  $\alpha_i \geq 0$  satisfying (33) and (35). Their aggregates

$$r_\pm = \sum_{\sigma_i = \pm 1} \alpha_i b_i$$

have  $r_+ + r_- = x - y$ . Since  $\sum_{\sigma_i = \pm 1} \alpha_i = (p \pm a)/2 = A_\pm$ , they satisfy  $\|r_\pm\| \leq A_\pm$  and hence  $r_+ - r_- \in D_0$ . No choice of these representations across different  $u$  or  $\delta$  is required.

For  $b \in Q$  and  $\sigma = \pm 1$ , evenness of  $\lambda$  gives

$$\lambda(\langle (b, \sigma), X_\delta \rangle) = \lambda(1 - \delta(1 - \sigma \langle b, u \rangle)).$$

The factors  $1 - \sigma \langle b, u \rangle$  belong to  $[1 - p_K(u), 1 + p_K(u)]$ , a compact subinterval of  $(0, \infty)$ . Lemma 3.2 therefore gives an error  $\varepsilon_u(\delta) \rightarrow 0$  such that

$$\left| \frac{\lambda(\langle (b, \sigma), X_\delta \rangle)}{\lambda(1 - \delta)} - 1 + \sigma \langle b, u \rangle \right| \leq \varepsilon_u(\delta) \quad (b \in Q, \sigma = \pm 1). \quad (44)$$

In particular, the normalized cost of the optimal representation is at least

$$p - \langle r_+ - r_-, u \rangle - p\varepsilon_u(\delta) \geq p - h_{D_0}(u) - p\varepsilon_u(\delta).$$

Compare this cost, using (35), with the two-term representation

$$(x - y, a) = p_1(x/p_1, 1) + p_2(-y/p_2, -1).$$

Both rows belong to  $Q \times \{-1, 1\}$ . By (44), their normalized cost tends to  $s - \langle x + y, u \rangle$ . Letting  $\delta \downarrow 0$  yields

$$\langle x + y, u \rangle \leq h_{D_0}(u) + s - p \quad (u \in \text{int } K). \quad (45)$$

The error was uniform only over  $b, \sigma$  for the fixed  $u$ ; that is all this limit requires.

**Separation and the median.** Continuity extends (45) to  $u \in K$ . For any nonzero  $v \in \mathbb{R}^n$ , the point  $u = v/h_Q(v)$  lies on  $\partial K$ , by (2). Homogeneity of support functions gives

$$\langle x + y, v \rangle \leq h_{D_0}(v) + (s - p)h_Q(v).$$

The same inequality holds at  $v = 0$ . The right side is the support function of the compact convex set  $D_0 + (s - p)Q$ , with the convention  $0Q = \{0\}$ . Convex separation therefore implies

$$x + y \in D_0 + (s - p)Q. \quad (46)$$

Choose the corresponding pair  $r_+, r_-$  in (43), and put

$$m = x - r_+ = y + r_-.$$

Then  $2m = x + y - (r_+ - r_-)$ , so (46) gives

$$\|m\| \leq c, \quad \|x - m\| \leq A_+, \quad \|y - m\| \leq A_-.$$

The triangle inequality forces all three bounds to be equalities:

$$\begin{aligned} p_1 &\leq \|m\| + \|x - m\| \leq c + A_+ = p_1, \\ p_2 &\leq \|m\| + \|y - m\| \leq c + A_- = p_2. \end{aligned}$$

Thus  $m$  lies between  $0, x$  and between  $0, y$ , and

$$\|x - m\| + \|m - y\| = A_+ + A_- = p = \|x - y\|.$$

This proves the median property. The argument permits  $c = 0$  or either  $A_\pm = 0$ , so saturated triangle inequalities require no limiting argument.  $\square$

## 8.1 Three balls and the equality classification

A normed space has the *three-ball intersection property*, also called the *3.2 intersection property*, if any three closed balls that intersect pairwise have a common point. Their radii may differ. The equivalence between this intersection property and the metric-median condition is Lima's classical norm-additive decomposition criterion [19, Theorem 1.2]; see also Hansen–Lima [13, Theorem 3.6(4)]. We include the short direction needed here, with arbitrary radii.

**Lemma 8.2** (From medians to three balls). *Suppose every triple of points in a real normed space has a metric median. Then the space has the three-ball intersection property, including balls of radius zero.*

*Proof.* Consider pairwise-intersecting balls with centers  $x_1, x_2, x_3$  and radii  $r_1, r_2, r_3 \geq 0$ . Let  $m$  be a metric median of their centers, and set  $d_i = \|m - x_i\|$ . Pairwise intersection and the median property imply

$$d_i + d_j = \|x_i - x_j\| \leq r_i + r_j \quad (i \neq j).$$

If every  $d_i \leq r_i$ , the median is a common point. Otherwise, after relabeling,  $t = d_1 - r_1 > 0$ . For  $j = 2, 3$  the displayed inequalities give  $d_j + t \leq r_j$ , so no other ball excludes  $m$ . Since  $0 < t \leq d_1$ , the point

$$z = m + \frac{t}{d_1}(x_1 - m)$$

is well defined and lies on the segment from  $m$  to  $x_1$ . It satisfies

$$\begin{aligned} \|z - x_1\| &= d_1 - t = r_1, \\ \|z - x_j\| &\leq \|z - m\| + d_j = t + d_j \leq r_j \quad (j = 2, 3). \end{aligned}$$

Thus  $z$  lies in all three balls. This includes  $r_1 = 0$ , when  $z = x_1$ . □

The required structure theorem is due to Hansen and Lima. In their terminology the balls are closed and have arbitrary radii [13, Section 1].

**Theorem 8.3** (Hansen–Lima). *A nonzero finite-dimensional real Banach space with the 3.2 intersection property is linearly isometric to a space obtained from the real line by finitely many  $\ell_1$  and  $\ell_\infty$  direct sums.*

This is [13, Corollary 7.4]. The two direct-sum norms are exactly those in (4), so their unit balls are Hanner polytopes.

**Proposition 8.4** (Classification of equality). *If an origin-symmetric convex body  $K \subset \mathbb{R}^n$ ,  $n \geq 1$ , satisfies  $P(K) = 4^n/n!$ , then it is a linear Hanner body.*

*Proof.* Put  $Q = K^\circ$ . Proposition 8.1 and Lemma 8.2 give the 3.2 intersection property for  $(\mathbb{R}^n, p_Q)$ . This is a real Banach space, since every finite-dimensional normed space is complete. Theorem 8.3 gives an invertible linear map  $T$  and a Hanner polytope  $H$  with  $Q = TH$ . By (3) and Lemma 2.2,

$$K = Q^\circ = T^{-\top} H^\circ$$

is a linear Hanner body. The converse was proved in Lemma 2.2; together with Corollary 6.2, this completes Theorem 1.1. □

## References

- [1] Bo Berndtsson. Complex integrals and Kuperberg’s proof of the Bourgain–Milman theorem. *Advances in Mathematics*, 388:107927, 2021. doi: 10.1016/j.aim.2021.107927. URL <https://arxiv.org/abs/2008.00838>.
- [2] Len Bos, Jean-Paul Calvi, and Norman Levenberg. On the Siciak extremal function for real compact convex sets. *Arkiv för Matematik*, 39:245–262, 2001. URL [https://archive.ymsc.tsinghua.edu.cn/pacm\\_download/116/7771-11512\\_2006\\_Article\\_BF02384556.pdf](https://archive.ymsc.tsinghua.edu.cn/pacm_download/116/7771-11512_2006_Article_BF02384556.pdf).
- [3] Jean Bourgain and Vitali D. Milman. New volume ratio properties for convex symmetric bodies in  $\mathbb{R}^n$ . *Inventiones Mathematicae*, 88(2):319–340, 1987. doi: 10.1007/BF01388911. URL <https://link.springer.com/article/10.1007/BF01388911>.
- [4] Shibing Chen, Yuanyuan Li, Dongmeng Xi, and Zhe-Feng Xu. The Mahler conjecture in three dimensions, 2026. URL <https://arxiv.org/abs/2605.09334v3>. Version 3, 15 June 2026.
- [5] Dario Cordero-Erausquin and Bo’az Klartag. Moment measures. *Journal of Functional Analysis*, 268(12):3834–3866, 2015. doi: 10.1016/j.jfa.2015.04.001. URL <https://doi.org/10.1016/j.jfa.2015.04.001>.
- [6] Jean-Pierre Demailly. Monge–Ampère operators, Lelong numbers and intersection theory. In Vincenzo Ancona and Alessandro Silva, editors, *Complex Analysis and Geometry*, University Series in Mathematics, pages 115–193. Plenum Press, New York, 1993. doi: 10.1007/978-1-4757-9771-8\_4. URL [https://link.springer.com/chapter/10.1007/978-1-4757-9771-8\\_4](https://link.springer.com/chapter/10.1007/978-1-4757-9771-8_4).
- [7] Matthieu Fradelizi and Mathieu Meyer. Some functional inverse Santaló inequalities. *Advances in Mathematics*, 218(5):1430–1452, 2008. doi: 10.1016/j.aim.2008.03.013. URL <https://doi.org/10.1016/j.aim.2008.03.013>.
- [8] Matthieu Fradelizi, Nathael Gozlan, and Simon Zugmeyer. Transport proofs of some functional inverse Santaló inequalities, 2021. arXiv:2109.00871v3. [Manuscript PDF](#).
- [9] Matthieu Fradelizi, Alfredo Hubard, Mathieu Meyer, Edgardo Roldán-Pensado, and Artem Zvavitch. Equipartitions and Mahler volumes of symmetric convex bodies. *American Journal of Mathematics*, 144(5):1201–1219, 2022. doi: 10.1353/ajm.2022.0027. URL <https://arxiv.org/abs/1904.10765>.
- [10] Yehoram Gordon, Mathieu Meyer, and Shlomo Reisner. Zonoids with minimal volume-product—a new proof. *Proceedings of the American Mathematical Society*, 104(1):273–276, 1988. doi: 10.1090/S0002-9939-1988-0958082-9.
- [11] Renan Gross. A conformal Skorokhod embedding. *Electronic Communications in Probability*, 24, 2019. doi: 10.1214/19-ECP272. URL <https://arxiv.org/abs/1905.00852>. Article no. 68, 11 pp.
- [12] Olof Hanner. Intersections of translates of convex bodies. *Mathematica Scandinavica*, 4:65–87, 1956. doi: 10.7146/math.scand.a-10456. URL <https://journals.msp.org/mscand/article/view/2863>.
- [13] Allan B. Hansen and Åsvald Lima. The structure of finite dimensional Banach spaces with the 3.2. intersection property. *Acta Mathematica*, 146:1–23, 1981. doi: 10.1007/BF02392457. URL <https://doi.org/10.1007/BF02392457>.

- [14] Hiroshi Iriyeh and Masataka Shibata. Symmetric Mahler’s conjecture for the volume product in the 3-dimensional case. *Duke Mathematical Journal*, 169(6):1077–1134, 2020. doi: 10.1215/00127094-2019-0072. URL <https://arxiv.org/abs/1706.01749v3>.
- [15] Roman Karasev. Mahler’s conjecture for some hyperplane sections. *Israel Journal of Mathematics*, 241:795–815, 2021. doi: 10.1007/s11856-021-2114-4. URL <https://arxiv.org/abs/1902.08971>.
- [16] Jaegil Kim. Minimal volume product near Hanner polytopes. *Journal of Functional Analysis*, 266(4):2360–2402, 2014. doi: 10.1016/j.jfa.2013.08.008. URL <https://arxiv.org/abs/1212.2544v1>.
- [17] Jaegil Kim and Artem Zvavitch. Stability of the reverse Blaschke–Santaló inequality for unconditional convex bodies. *Proceedings of the American Mathematical Society*, 143(4):1705–1717, 2015. doi: 10.1090/S0002-9939-2014-12334-3. URL <https://arxiv.org/abs/1302.5719v1>.
- [18] Greg Kuperberg. From the Mahler conjecture to Gauss linking integrals. *Geometric and Functional Analysis*, 18(3):870–892, 2008. doi: 10.1007/s00039-008-0669-4. URL <https://arxiv.org/abs/math/0610904>.
- [19] Åsvald Lima. Intersection properties of balls in spaces of compact operators. *Annales de l’Institut Fourier*, 28(3):35–65, 1978. doi: 10.5802/aif.700. URL <https://aif.centre-mersenne.org/articles/10.5802/aif.700/>.
- [20] Magnus Lundin. The extremal PSH for the complement of convex, symmetric subsets of  $\mathbb{R}^N$ . *Michigan Mathematical Journal*, 32(2):197–201, 1985. doi: 10.1307/mmj/1029003186.
- [21] Kurt Mahler. Ein Minimalproblem für konvexe Polygone. *Mathematica (Zutphen)*, B7:118–127, 1938.
- [22] Kurt Mahler. Ein Übertragungsprinzip für konvexe Körper. *Časopis pro pěstování matematiky a fysiky*, 68(3–4):93–102, 1939. doi: 10.21136/CPMF.1939.109441. URL <https://dml.cz/handle/10338.dmlcz/109441>.
- [23] Mathieu Meyer. Une caractérisation volumique de certains espaces normés de dimension finie. *Israel Journal of Mathematics*, 55(3):317–326, 1986. doi: 10.1007/BF02765029. URL <https://link.springer.com/article/10.1007/BF02765029>.
- [24] Fedor Nazarov. The Hörmander proof of the Bourgain–Milman theorem. In Bo’az Klartag, Shazar Mendelson, and Vitali D. Milman, editors, *Geometric Aspects of Functional Analysis*, volume 2050 of *Lecture Notes in Mathematics*, pages 335–343. Springer, Berlin, Heidelberg, 2012. doi: 10.1007/978-3-642-29849-3\_20. URL [https://doi.org/10.1007/978-3-642-29849-3\\_20](https://doi.org/10.1007/978-3-642-29849-3_20).
- [25] Fedor Nazarov, Fedor Petrov, Dmitry Ryabogin, and Artem Zvavitch. A remark on the Mahler conjecture: local minimality of the unit cube. *Duke Mathematical Journal*, 154(3):419–430, 2010. doi: 10.1215/00127094-2010-042. URL <https://arxiv.org/abs/0905.0867>.
- [26] OpenAI. The Mahler Conjecture for General Convex Bodies. OpenAI Math Release preprint OAI:The-Mahler-Conjecture-for-General-Convex-Bodies-September-22-2026, 2026.

- [27] Shlomo Reisner. Random polytopes and the volume-product of symmetric convex bodies. *Mathematica Scandinavica*, 57:386–392, 1985. doi: 10.7146/math.scand.a-12124. URL <https://journals.msp.org/mscand/article/view/1544>.
- [28] Shlomo Reisner. Zonoids with minimal volume-product. *Mathematische Zeitschrift*, 192(3): 339–346, 1986. doi: 10.1007/BF01164009.
- [29] Shlomo Reisner. Minimal volume-product in Banach spaces with a 1-unconditional basis. *Journal of the London Mathematical Society*, 36(1):126–136, 1987. doi: 10.1112/jlms/s2-36.1.126. URL <https://doi.org/10.1112/jlms/s2-36.1.126>.
- [30] Jean Saint-Raymond. Sur le volume des corps convexes symétriques. In *Séminaire d’initiation à l’analyse, 20e année, 1980–1981*. Université Paris VI, Paris, 1981. Exposé no. 11.